BISHNU PRASAD KAR

AI/ML Engineer – Machine Learning, LLMs, Agentic AI

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SUMMARY

AI/ML engineer and Computer Science undergraduate (B.Tech, 2027). Fine-tuned transformers to 99.95% accuracy on 43K+ documents, built a five-agent LLM system with verifiable decision trails, and shipped inference services with FastAPI and Docker. Three conference papers presented.

TECHNICAL SKILLS

Programming & Data: Python, SQL, Bash, pandas, NumPy, Matplotlib, Tableau

Machine Learning: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, NLP, Computer Vision, SHAP, model evaluation

Generative AI: LLMs, Agentic AI, RAG, LangChain, FAISS, prompt engineering, llama.cpp, Ollama, Gemini API

Systems & Data Platforms: FastAPI, Flask, REST APIs, PostgreSQL, MySQL, Hadoop, Spark, Docker, Git, Linux

Quantum & Ledger: Qiskit, Qiskit Aer, QAOA, quantum kernels, Hyperledger Fabric, Solidity

EXPERIENCE

Data Science Intern, Jyesta Corporate Entity x E-Cell, IIT Roorkee Dec 2025 - Feb 2026

- Built recommendation and loan-prediction systems with feature engineering, multiple classifiers and model explainability.

Trainee: Generative AI, Data Analytics, Cloud Computing, Central Tool Room & Training Centre (CTTC), Ministry of MSME 2024 - 2026

- Generative AI (2026): deep learning, LLMs, Hugging Face, LangChain, RAG, end-to-end capstone. Data Analytics (2025): Python, SQL, Tableau.

Backend Engineering Trainee, Boot.dev (Remote) Jul 2025 - Jul 2026

- Completed a full backend track: Python, SQL, Linux, Docker, networking, system design.

PROJECTS

Q-CONSENSUS – Multi-agent LLM reasoning with verifiable decision trails 2026

Python, FastAPI, Qiskit Aer, llama.cpp, Solidity, Docker

- Five agents debate on a local LLM; a QAOA solver over quantum-kernel similarity picks the consensus answer, benchmarked against a classical baseline.
- SHA-256 hash-chained events anchored on a private Ethereum chain, so any run can be replayed and verified.

Misinformation Vaccine – Fake news detection, URL to verdict 2025

Python, PyTorch, Transformers, scikit-learn, FastAPI, PostgreSQL

- DistilBERT fine-tuned on 43,944 articles: 99.95% accuracy and 0.9995 F1 against a 98.84% TF-IDF baseline.
- Scraping, a FastAPI inference service and stored history; dataset-bias limits documented.

Floodwatch – IoT flood early-warning; 1st position, Model Fiesta, Suravi 2K26 2026

Python, scikit-learn, SHAP, FastAPI, Ollama, Arduino

- 14 engineered features from three sensors feed a four-level risk model with SHAP attributions.
- A local LLM explains high-risk readings and the system triggers a hardware alert.

Also: QuantumLab (Qiskit Aer lab); Soil-to-Shelf (Hyperledger Fabric traceability); CycleSense (IoT and LLM health); ShelfSight (OpenCV and OCR scanning).

EDUCATION

B.Tech in Computer Science and Engineering, Gandhi Institute of Excellent Technocrats, Bhubaneswar Expected Aug 2027

CGPA 8.8/10 (through semester 6). Coursework: Machine Learning, Artificial Intelligence, Data Structures and Algorithms, Database Systems.

PUBLICATIONS

Entropy-Optimized Adaptive Hybrid Thresholding for Retinal Blood Vessel Segmentation. *SPARK-2K26 Research Conclave, GIET, Jan 2026*

CityFlow: A Scalable Big Data Framework for Real-Time Analytics in Computing 4.0. *NCCENGT-2025, Sep 2025*

A Modular and Scalable Arithmetic and Logical Unit Design through Multiplexer-Based Controller. *ICSTRE-2024, Nov 2024*

ACHIEVEMENTS

1st position, Model Fiesta, Suravi 2K26, GIET Bhubaneswar (Feb 2026), for the Floodwatch flood early-warning system; 3rd place, ReFlow Menstrual Health Innovation Hackathon, IIT Bhubaneswar (Nov 2025); top performer, 8-week Machine Learning training, Internshala, 94% in the final assessment (2025); Completion with Merit, one-month Quantum Computing course, C-DAC, 81.67%.

CERTIFICATIONS

Getting Started with Deep Learning (NVIDIA DLI, 2025); CS50's Introduction to AI with Python (Harvard, 2025); Career Essentials in Generative AI (Microsoft and LinkedIn, 2024); Quantum Computing with Merit, 81.67% (C-DAC); Big Data specialisation, four courses (UC San Diego, 2025).